



ADELEKE KEHINDE BOLUWATIFE

SOFTWARE ENGINEER
AI & FULL STACK SYSTEMS

CONTACT

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- Remote ready · Worldwide

PROFILE SNAPSHOT

Backend & APIs	Expert
AI Agents & LLM Systems	Expert
Enterprise ERP / Django	Advanced
Frontend Product	Advanced
Data & Databases	Advanced
DevOps & Delivery	Proficient

WHAT I BRING

- Ships to production with payments, auth, approvals and real users, not demos.
- Owns the full stack, from database schema to interface motion.
- Builds AI that is safe in the enterprise: permissions enforced in code, humans approve every write.
- Turns spreadsheets into systems that non technical teams actually adopt.
- Comfortable inside enterprise process: protected branches, reviews, CI, UAT.

"Every platform I build is judged by one question: is it live, and is someone depending on it?"

Adeleke Kehinde Boluwatife

Software Engineer · Applied AI & Multi Agent Systems · Backend & ERP

Software engineer with **3+ years of continuous professional delivery**, beginning while still completing my B.Sc at Caleb University and running unbroken since. I am currently a **Software Engineer at Hitech Construction Company**, building the AI and operations layer of an enterprise construction ERP: a **department scoped multi agent system**, the **Paytrack HRIS**, and the **Survey** and **OHS** modules. In parallel I lead client platform delivery at **THCO**, with systems in daily use at IHS Towers, Talent Nation and TASCK. I design and ship **production platforms end to end**: Django, .NET, FastAPI and NestJS backends, React and Next.js frontends, and LLM pipelines that produce real output for real users. I care about the parts that are hard to fake: correct data models, permissions enforced in code, clean API contracts, and software that survives contact with real users.

3+ YEARS PROFESSIONAL	10 PRODUCTION SYSTEMS SHIPPED	9 AI AGENTS IN THE ERP	700+ AUTOMATED TESTS ON AI LAYER
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PROFESSIONAL EXPERIENCE

Software Engineer

CURRENT ROLE

Jul 2026 to Present

Hitech Construction Company Limited · Full time · Enterprise ERP & Internal Systems · Lagos

- Joined the ERP engineering team of one of Nigeria's largest civil construction firms, working on a **30+ app Django ERP** (PostgreSQL, Celery, DRF) used across HSE, HR, Survey, Quantity Surveying, Fleet, Cost Control and Drone Mapping.
- Within three months, delivered the company's **AI agent layer**, HRIS modules on **Paytrack**, the **Survey inventory dashboard and flow** and the **OHS reporting flow**, and took ownership of the **biometric attendance platform**.

APPLIED AI · MULTI AGENT · RAG

Hitech ERP AI Agent Layer

8+1 DEPT AGENTS + ORCHESTRATOR	2 PATH RAG	700+ TESTS	0 UNAPPROVED WRITES
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- Designed and built a **multi agent layer over the ERP**: HSE, QS, BOQ, HR, Fleet, Drone, Cost Control and Survey agents, coordinated by an **orchestrator** that routes questions, runs bounded agent to agent conversations and reconciles conflicts, exposed through per department chatbots.
- Scope enforced in code, not in the prompt**: every read resolves against a department registry before any SQL runs, and executes as the requesting user through the ERP's own permission functions. A prompt injected request for payroll from the HSE agent fails at the tool boundary.
- Agents never write**. They raise proposed changes that apply only after the ERP's approval workflow signs them off, with every action in an append only audit log. The orchestrator holds no data tools at all.

BIOMETRIC ATTENDANCE

ZKBio Time Platform Owner

- Own Hitech's biometric attendance server: **28 face recognition terminals** and **12,000+ employees**. Mapped its REST API, audited its configuration, and designed **day and night auto shift scheduling** from real punch data for a staged site pilot.

ZKBio Time · REST API · Attendance & Shift Design

HITECH AT A GLANCE

ERP apps in scope	30+
Department AI agents	8
Orchestrator	1
AI layer tests	700+
Survey sites covered	30
Survey permissions	47
Biometric terminals	28
Employees on attendance	12,000+

AI DESIGN PRINCIPLES

- **Permissions in code.** A tool that refuses an out of scope model is a boundary; an instruction is a suggestion.
- **Humans approve writes.** AI proposes, people decide, the audit log remembers.
- **Counts come from SQL.** Tabular data is queried, never guessed from retrieved chunks.
- **Sensitive by default.** Pay, bank and private fields are masked until an admin releases them.
- **Reuse, don't reinvent.** The ERP's approval, audit and access systems are the AI's too.

STACK AT HITECH

Django 5

DRF

Celery

PostgreSQL

pgvector

HTMX

Bootstrap 5

Claude API

OpenAI API

Llama

RAG

REST APIs

ZKBio Time

Excel Import Pipelines

Hitech Construction Company / continued

APPLIED AI · CONTINUED

AI Agent Layer: Knowledge, Privacy & Data Linking

- > **Dual path RAG.** Narrative documents are chunked and embedded into **pgvector** inside the existing PostgreSQL database; spreadsheets are parsed into typed rows and answered with **real SQL**, so “how many excavators do we have” returns a count, not a guess.
- > **Restricted detail disclosure loop.** Answers automatically mask pay, bank and private columns; a reader can request them, an administrator approves or declines, and approved values are read fresh from source and delivered into the same conversation.
- > **Project identity reconciliation.** Harvests every way departments name the same project (Candy, Drone, Wialon fleet units, Survey) and proposes the canonical match for a human to confirm, remembering rejections so it never asks twice.
- > **Provider abstraction** across Claude, OpenAI and Llama with privacy aware fallback, key monitoring and usage tracking, built for a **sovereign, on premise** deployment path. Also: streaming answers, deep links into ERP records, follow up suggestions and per user feedback.

Django · Celery · PostgreSQL · pgvector · Claude API · RAG · Embeddings · Multi Agent Orchestration

SURVEY DEPARTMENT · DASHBOARD & FLOW

Survey Inventory & Asset Management

- > Replaced two shared Excel workbooks (**16 worksheets, 30 sites**, 33 placed surveyors) with a full ERP module: equipment register, **issue and return custody**, maintenance and repairs, calibration, consumables stock book, deployment board, vehicles, software licences, store requests, stock take and reports.
- > Built an **Excel import pipeline** that canonicalises one site written six different ways and infers equipment condition from free text, preserving anything it cannot parse so nothing is lost.
- > **Stock balances are derived from a movement ledger**, so the number on screen can never disagree with the entries; expired licences and low stock are flagged automatically.
- > Access control enforced in **three independent layers** (navigation, route guards, site data scoping) across **six roles and 47 permissions**, meeting the Survey Manager's brief that the team sees only what it needs.

Django · PostgreSQL · Import Pipelines · RBAC · Dashboards

HR & WORKFORCE

Paytrack HRIS

- > Built HRIS modules on Hitech's payroll platform: **notifications, weekly site reports** and **role based dashboards**.
- > **ID card lifecycle tracking:** print, dispatch, receive and deliver, so every card is accounted for.
- > Extended the shared staff request and approval engine instead of building parallel models, inside a team branch and review workflow.

Django 5.2 · HTMX · Bootstrap 5 · PostgreSQL

HEALTH & SAFETY

OHS Reporting & Dashboard

- > Built the **Occupational Health & Safety** flow: incident and injury logs, weekly and monthly medical reports and the **monthly OHS report**.
- > Live **OHS dashboard** with per site scoping, checklist templates, document numbering and report templates replacing Excel based HSE and ESG reporting.

Django · Report Templates · Dashboards

TECHNICAL SKILLS

LANGUAGES

Python C# TypeScript
JavaScript SQL HTML / CSS

BACKEND

Django DRF Celery
ASP.NET Core FastAPI NestJS
Node.js REST API Design
Prisma Auth / JWT Swagger

FRONTEND

React 19 Next.js 15 HTMX
Vite Tailwind CSS Bootstrap 5
Design Systems Framer Motion

AI / ML

Multi Agent Systems Claude API
OpenAI API LangChain
LangGraph RAG Embeddings
Prompt Eng. Fine tuning
Scikit Learn Hugging Face

DATA

PostgreSQL pgvector
SQL Server MongoDB Redis
Oracle DB SQLite ChromaDB
FAISS Pinecone

DELIVERY & TOOLING

Git / GitHub Azure DevOps
Docker CI Pipelines Vitest
xUnit Postman Vercel
AWS basics

ENGINEERING PRACTICE

- Protected branch workflows with PR review and staged environments.
- Feature flagged rollouts and staged client demos.
- Domain modelling before code, so schema decisions are made once, correctly.
- Handover docs, PRDs and API contracts treated as deliverables.

PROFESSIONAL EXPERIENCE / CONTINUED

Software Engineer **CONCURRENT**

Mar 2025 to Present

THCO · Client Platform Delivery · Lagos / Remote

- Lead engineer on **four concurrent client platforms**, taking each from first stakeholder call through architecture, build, demo loop and production deployment.
- Ran **weekly client demo cycles** with founders and enterprise stakeholders, translating ambiguous product intent into scoped, shippable engineering increments.

TASCK OS for MI Agbaga

"Creativity, Managed." An operating system for Africa's creative economy: admin command centre with CRM pipeline, scraper agents and meeting intelligence, plus brand, creative and super creative portals covering briefs, deliverables, approvals, invoices and wallets.

Talent Nation for Iyinoluwa Aboyeji & Kesiah Valkin

AI Engineering Fellowship platform: cognitive assessment games, a seven day readiness sprint, mission pools, knowledge trees, placement and remediation flows, campus kiosk devices and an admin console, all on React/Vite frontends over a NestJS, PostgreSQL and Redis backend.

IHS Procure for the procurement team, IHS Towers

Designed and built the **vendor onboarding user flow system** for one of Africa's largest telecom infrastructure companies: a gated multi step compliance wizard (declarations, categories, supplier forms, documents, licences, risk scoring) feeding a live contracts, tenders and performance dashboard.

Oyela for Mr Praise

Culturally aware family systems assessment platform converting guided diagnostics into narrative AI coaching reports across marital, premarital, parenting, teen and executive pathways, with payment integration, coach marketplace and templated PDF report generation.

React Next.js 15 TypeScript NestJS FastAPI PostgreSQL MongoDB Redis
Claude AI Docker Azure DevOps

Backend Engineer

Feb 2026 to Present

Larkwave · Contract · Concurrent Engagement

- Own the **entire backend surface of a multi tenant Point of Sale platform** built on **C# / ASP.NET Core**, authoring every endpoint across authentication, terminals, catalogue, inventory, orders, settlement, reporting and notifications.
- Designed the **terminal authentication model** (device pairing, scoped tokens, configurable refresh windows) so physical POS terminals stay session safe in low connectivity retail environments.
- Built **transactional messaging infrastructure**: contact message and newsletter APIs over a hardened HTTPS email provider integration with fail fast delivery semantics and structured error surfacing.
- Standardised **API documentation with live Swagger UI across every environment**, cutting frontend integration turnaround and removing ambiguity from contract handover.
- Engineered with an **AI assisted workflow**, using LLM tooling to accelerate scaffolding, test generation and refactors while retaining full architectural ownership and review of shipped code.

C# / .NET ASP.NET Core REST API Design SQL Server Swagger / OpenAPI
Auth & Tokens AI Assisted Engineering

EDUCATION

Caleb University, Lagos
B.Sc Computer Science
2020 to 2024 · GPA 4.4 / 5.0 (Distinction)

FSTC Ijebu Imushin
WAEC, Senior Secondary
2013 to 2019

CONTINUING EDUCATION

- DeepLearning.AI, LLM application development & agentic systems.
- Hugging Face, transformers, embeddings and model fine tuning.
- Self directed study in LangChain, LangGraph and production RAG design.

INDUSTRIES SERVED

- **Construction:** ERP, HSE, HR, survey assets
- **Telecoms:** enterprise procurement
- **Government:** communications regulation
- **Creative economy:** CRM & operations
- **Edtech:** assessment & placement
- **Retail:** point of sale
- **Healthcare AI:** diagnostic datasets

PROFESSIONAL EXPERIENCE / CONTINUED

Data Annotation Specialist

Nov 2024 to Feb 2025

Awarri · Contract

- Key contributor on a **high precision computer vision programme for healthcare diagnostics**, producing pixel perfect semantic segmentation and object detection data across complex hospital laboratory environments.
- Executed classification, entity recognition and relevance labelling for production ML models, raising dataset reliability through guideline compliant workflows and peer QA.

Computer Vision

Semantic Segmentation

NLP Labelling

ML Datasets

AI Developer

Jan 2024 to Oct 2024

Nigerian Communications Commission (NCC) · Full time

- Spearheaded the integration of AI into **national regulatory frameworks**, building ML models that automated compliance review tasks and measurably improved operational throughput.
- Designed and deployed **RAG powered document intelligence** using LangChain, FastAPI and vector databases for retrieval across large regulatory corpora.
- Led prompt engineering initiatives that **improved LLM response accuracy by 40%** on regulatory document analysis tasks.
- Collaborated with cross functional digital transformation teams to move prototypes into supported internal tooling.

Python

LangChain

RAG

FastAPI

Vector DBs

Prompt Engineering

AI & Full Stack Developer

Feb 2023 to Dec 2023

Independent / Freelance · While completing B.Sc, Caleb University

- Started professional delivery **while still an undergraduate**, building intelligent applications for small businesses using LLMs, RAG and modern web stacks.
- Shipped multilingual chatbots, retail intelligence tooling and document analysis assistants against real client requirements and budgets.
- Established the working pattern I still use: scope, architect, demo weekly, deploy, then support.

Python

FastAPI

React

OpenAI API

Client Delivery

Database Manager / Administrator

Apr 2022 to Jan 2023

Sonet Technology Limited · SIWES Industrial Training

- Managed and tuned Oracle database performance, delivering a **20% improvement in query response times** through index and query optimisation.
- Implemented backup and recovery procedures maintaining **100% data integrity** across system upgrades and migrations.

Oracle DB

SQL Tuning

Backup & Recovery

HOW I RUN AN ENGAGEMENT

STEP 01

Discover

Stakeholder calls, domain modelling and honest technical scoping before a line of code.

STEP 02

Architect

System design, stack selection and data modelling with production constraints from day one.

STEP 03

Build & Iterate

Weekly demo loops, staged rollouts behind feature flags, tests where they pay off.

STEP 04

Ship & Operate

Deployment, monitoring, handover docs and the follow through that keeps platforms alive.

◆ LANGUAGES

- English: Fluent
- Yoruba: Native
- French: Basic

◆ INTERESTS

- Applied AI research & LLM system design.
- Open source contribution.
- Teaching, mentoring & technical writing.
- Product design & typography.

◆ FIND MY WORK

- ✓ adeleke-kehinde-portfolio.vercel.app
- 🔗 github.com/hexkoda23
- ✉ kehindeadeleke92@gmail.com

References and client introductions available on request.

ENGINEERING & AI PROJECTS

NLP · TRANSFORMERS

AI Résumé & Job Match Scoring

Scores CVs against job descriptions using transformer embeddings and cosine similarity, returning ranked skill gap explanations rather than an opaque number.

Python · FastAPI · Sentence Transformers

AGENTIC RAG

NYSC AI Chatbot

Three tool agent architecture combining local vector search with live retrieval, answering programme questions with grounded, source attributed responses.

LangChain · LangGraph · FastAPI

BUSINESS INTELLIGENCE

AI Business Insight Generator

Analyses raw CSV business data to produce executive level narrative insights, detect anomalies and render interactive charts automatically.

Python · Pandas · OpenAI API · Plotly

UNSUPERVISED ML

AI Fraud & Anomaly Detection

Detects unusual transaction patterns with Isolation Forest and autoencoder reconstruction error, surfacing ranked risk cases for human review.

Scikit Learn · Isolation Forest · Autoencoders

ADAPTIVE AGENT

Intelligent AI Study Planner

Agent based learning system with long term memory in a vector store, rebuilding study plans as a learner's performance and availability change.

LLMs · Pinecone · FastAPI · React

FULL STACK RETAIL

Smart Provision Shop Manager

Retail management system with an AI chat assistant, real time inventory tracking, sales analytics and restock forecasting for small shop owners.

React · FastAPI · SQLite · OpenAI API

SELECTED ACHIEVEMENTS

- ▶ **Built a permission safe multi agent AI layer** for an enterprise construction ERP within my first three months at Hitech: nine agents, dual path RAG, human approval on every write, 700+ automated tests.
- ▶ **Replaced spreadsheet operations with live ERP modules** for Hitech's Survey, HSE and HR teams, covering equipment custody across 30 sites, OHS reporting and HRIS workflows.
- ▶ **Shipped six client platforms to production** across enterprise procurement, edtech, creative CRM, retail POS and AI assessment. Every engagement delivered, none abandoned at prototype.
- ▶ **Sole backend author of a live Point of Sale platform** in C# / ASP.NET Core, covering authentication, terminals, inventory, orders, settlement and reporting.
- ▶ **Trusted with an enterprise account**, building the vendor onboarding user flow system for IHS Towers' procurement team inside their Azure DevOps engineering process.
- ▶ **Improved LLM response accuracy by 40%** at the Nigerian Communications Commission through systematic prompt engineering and retrieval design.
- ▶ **Graduated with distinction:** B.Sc Computer Science, GPA 4.4 / 5.0, Caleb University (2024), while already working professionally.

Open to Senior Engineering & AI Opportunities

Interested in **backend, full stack and applied AI engineering roles**, AI consulting and ambitious product builds. Based in Lagos, Nigeria, and **remote ready worldwide**.

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